REVIEW

An infectious diseases perspective on the microbiome and allogeneic stem cell transplant

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Purpose of review

The gut microbiome presents a novel source of diagnostic and therapeutic potential to modify post allogeneic stem cell transplant complications. There is an explosion of interest in microbiome research, mostly in the form of single-centre prospective time-series cohorts utilizing a variety of sampling frequencies and metagenomic technologies to sequence the microbiome. The purpose of this review is to summarize important recent publications and contextualize them within what has already been described in this rapidly growing field.

Recent finding

Results from observational human cohort and animal transplant models add to the growing body of evidence that the microbiome modulates the immunopathogenesis of posttransplant complications. This is particularly the case for recipients of grafts replete with T cells where the evidence that acute graft-versus-host disease is mediated by anaerobic commensal-associated short-chain fatty acids, which interact with mucosa-associated ($CD4^{+}FOXP3^{+}$) T-regulatory cells.

Summary

Future human research into the role of the microbiome in allogeneic stem transplant should incorporate rigorous and considered experimental design in addition to next-generation sequencing technology to better portray microbiome functional potential and active gene expression. In combination with host immune phenotyping, which would facilitate a robust understanding of the host–microbiome interaction that is required before meaningful translation into clinical diagnostics and therapeutics can be expected.

Keywords

alpha diversity, infection, microbiome, stem cell transplant

INTRODUCTION

It is now understood that the relationship between the intestinal microbiome and the innate and adaptive immune system in the gut determines whether the host is in homeostasis or one of a variety of well described disease states [1–4]. It is increasingly appreciated that outcomes after allogeneic stem cell transplant, can similarly be correlated with signatures in the gut microbiome. However, significant limitations in human microbiome experimental design and methodology confound findings and hinder rapid clinical translation of observations from such trials. A review of the recent literature in the context of stem cell transplant with a particular focus on animal models, infection and antibiotics will follow and concludes with a summary of a recommendation for future research directions. The intention is that this review will inform the clinician of the current state of knowledge about the role of the microbiome in transplant and highlight gaps that require further research.

APPROACH TO UNDERSTANDING THE LINKS BETWEEN THE MICROBIOME AND ALLOGENEIC STEM CELL TRANSPLANT

Study design adapted to microbiome studies

Peled *et al.* [5■■] interrogated the stool microbiome from 1362 recipients of allogeneic stem cell transplant from four transplant centres on three continents in the

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KEY POINTS

- The gut microbiome presents a novel source of diagnostic and therapeutic potential to modify post allogeneic stem cell transplant complications.
- Observational human cohort and animal transplant models add to the growing body of evidence that the microbiome modulates posttransplant immunopathogenesis.
- Rigorous and considered experimental design in combination with host immune phenotyping is required to facilitate a robust understanding of the host–microbiome interaction before meaningful translation into clinical diagnostics and therapeutics can be expected.

seminal prospective cohort study published in the *New England Journal of Medicine*. Using a discovery and validation study design, a total of 8767 stool samples (median of 4 per participant) were analysed using 16S ribosomal RNA sequencing. Integral to the strength of the study design was the centralized extraction, processing and sequencing of faecal samples at a single study laboratory and two control cohorts to which the study samples were compared. The results of this study, the largest prospectively collected cohort to date, replicate those observed in smaller single centres, whereby transplant is associated with enteric domination, loss of alpha diversity, an increased risk of subsequent transplant-related mortality and death attributed to GVHD, particularly in recipients of T-cell-replete grafts [6▪,7–11]. This provides the evidence to support the prior observations made in small cohorts and mouse models that the interaction between an injured microbiota and alloreactive mature T cells in the graft are required for GVHD immunopathogenesis. Although the strengths of this study are the rigorous multicentre study design and large number of included participants, the significant drawback is the 16S methodology, which, while cost effective, offers only a limited view of the microbiome constituents because of the inherent limitations in the metagenomic technology [12]. Further attempts to establish the mechanisms through which this alloreactivity is mediated, while balanced against cost, require more robust molecular methodology to undertake a functional assessment of the microbial genome and active gene expression in addition to profiling the host immunological response [12].

Improved sample frequency

Increasing the frequency of microbiome sampling in time-series cohorts is a way to improve the chance of detecting whether microbiome changes are the origin or the consequence of clinical outcome under investigation. The number of samples collected in microbiome studies may be limited for a variety of reasons including the inherent challenges related to collecting a specimen produced with variable frequency, the prohibitive costs of metagenomic technology and experimental infrastructure. However, multiple publications have demonstrated that there are important clinical insights to be gained by increasing the frequency of microbiome sampling. For example, Morjaria *et al.* [13▪] described a cohort of 18 patients undergoing either autologous and allogeneic haematopoietic cell transplant (HCT) who underwent intensive daily stool sampling during hospitalization. As well as 16S metagenomics, stool bacterial density with quantitative PCR and qualitative consistency measures were incorporated to assess the effect of antibiotics on the microbiota over the course of transplant. Day-to-day community shifts and compositional volatility of the microbiota demonstrated considerable individual patient variability in terms of the rapidity and magnitude of microbiota alteration in response to antibiotics, with the greatest shift occurring immediately post engraftment and in response to meropenem or piperacillin–tazobactam where up to greater than 99% of obligate anaerobes were lost in some patients. Similarly, Kusakabe *et al.* also undertook 16S rRNA sequencing of daily stool specimens commencing at day 8 before HCT conditioning until day 30 after HCT in a cohort of allogeneic and autologous HCT recipients and compared them to a control cohort [14]. Interestingly, analysis could divide recipients into those with a microbiota composition and diversity that drastically fluctuated over the course of HCT compared with those that maintained a robustly stable microbiota over the course of transplant, independent of the effects of traditionally identified microbiome confounders, such as antibiotic exposure, donor-type, conditioning, and so forth. Those patients who that lacked microbiota stability in addition to lacking diversity were associated with worse outcome. These findings require validation in additional cohorts but hint at potential interval censoring in historical cohorts where traditionally identified confounders, such as antibiotics typically precede pre-engraftment stool and may appear to have a stronger association with posttransplant outcomes.

Limiting microbiome confounders

The microbiome is susceptible to a wide range of external and environmental influences including geography, age, diet and nutritional status, BMI, ethnicity, medications, immune suppression and antibiotics [15]. Nutrition is increasingly associated with a significant influence on the microbiome. Outside of transplant populations, both murine and human studies have shown that enteral nutrition provides an intraluminal stimulus for optimal mucosal, barrier and immunological function of the gut [16]. Nutrition administered parenterally in mice results in significant shifts in the mouse enteric microbiome from a benign composition of mostly firmicutes to a predominantly Gram-negative, proteobacteria-dominated microbiome, which correlated with epithelial barrier dysfunction, a mucosal proinflammatory cytokine state and associated villous atrophy with increased risk of infection with enteric pathogens [17,18]. In humans, a trend to a greater abundance of beneficial SCFA producing anaerobic taxa was observed in those administered enteral compared to parenteral nutrition and preceded two recent studies in allogeneic HCT, one in a paediatric and the other an adult cohort, to investigate the association of peri transplant nutrition on the microbiome and posttransplant outcomes [11,19]. In one study a rigorous methodology of a pilot randomized trial of early enteral nutrition compared to standard care was undertaken (adult cohort) while in the paediatric cohort the methodology was less clear but a cohort of ten children were enrolled in an enteral nutrition arm followed by ten matched children in the parenteral therapy arm. In both cohorts enteral nutrition was associated with a significant recovery of diversity and greater abundance of short-chain fatty acid (SCFA)-producing obligate anaerobic taxa when compared with parenteral nutrition and physiologically restored luminal concentrations of SCFA only in those enterally fed [20,21▪] Whether enteral nutrition microbiome-manipulation strategies could present an opportunity to modify posttransplant outcome is a possibility but presents an appealing potential target in light of the well described association with poor outcomes and parenteral nutrition posttransplant, which may yet be explained by changes in the microbiome [22].

STUDIES ON THE MICROBIOME AND ALLOGENEIC HAEMATOPOIETIC CELL TRANSPLANT

Animal studies

Animal models provide supportive evidence that the microbiome interacts with the innate and adaptive immune system via the intestinal epithelium to regulate homeostasis. Commensal obligate anaerobes derived from the healthy human gut, particularly Clostridial species, mediate gut homeostasis in mice by upregulation of anti-inflammatory ($CD4^+FOXP3^+$) regulatory T cells (T_{reg}) and are in fact essential for the induction and maintenance of intestinal ($CD4^+FOXP3^+$) T_{regs} [23–25]. SCFA induce colonic regulatory ($CD4^+FOXP3^+$) T_{reg} and Th17 differentiation, and luminal concentrations correlate with number of colonic T_{reg}, which provides molecular insight into how the enteric microbiota mediates host–microbial homeostasis [26]. When mice are subjected to stem cell transplant and are then exposed to systemic antibiotics to induce a peri-engraftment loss of enteric alpha diversity with, expansion of enteric pathogens, and loss of commensal SCFA-producing anaerobes, they demonstrate increased rates of acute GVHD as do humans [8,10,26–29]. Associated with the depletion of resident butyrogenic microbiota, mice demonstrate a reduction in colonic T_{reg}, the effects of which can be partially ameliorated by either a re-introduction of commensal SCFA-producing enteric microbiota or direct application of SCFA into the gut lumen by oral gavage [8,27,30]. Animal models make up an important component of microbiome studies, particularly in the setting of allogeneic stem cell transplant in humans where significant confounders can dilute the ability to clearly define scientific correlations between microbiome and phenotype. In animal models, wherever variables can be controlled, these immunopathological pathways can be investigated further to gain more robust mechanistic insight.

Human studies

The therapeutic success of allogeneic stem cell transplant is currently limited by treatment-related complications and relapse of underlying disease, both of which have been associated with compositional change in the enteric microbiota in multiple series [5▪▪,31]. Typically, marked compositional changes occur with a reduction in alpha diversity, with loss of butyrogenic obligate anaerobes and domination by a single taxon or few taxa [5▪▪,6▪,7,8,11,13▪] (Table 1). Early single-centre cohort studies formed the foundation of this understanding including the original study by Taur *et al.* [32], which stratified peri-engraftment stool alpha diversity from 80 allogeneic stem cell recipients into high, medium and low diversity and found patients with low diversity were three times more likely to die within the 2-year study follow-up period compared with those with high diversity. In a cohort where donor and recipient stool pairs are compared, a trend toward greater phylogenetic distance between donor and recipient stool and higher grade of subsequent GVHD provide a roadmap for a future where novel diagnostic risk stratification for GVHD might incorporate measures

Table 1. Summary of studies and associated risk of subsequent graft-versus-host disease post allogeneic haematopoietic cell transplant

Alpha diversity	**Time of measurement**
Decrease associated with increased GVHD risk	Peri-engraftment [6■,8,9,33,36,37,38]
No association with GVHD	Peri-engraftment [20], 2 weeks prior to conditioning [34], immediately pre-conditioning [39]
Specific bacterial taxa, family or phyla	
Increased abundance with an increase GVHD	Lactobacillales [8], *Enterococcus* [35], Clostridales[35], *Fusobacterium* [38], *Firmicutes* [33,34], *Actinobacteria* [33], Enterobacteriaceae [6■]
Decreased abundance with increased GVHD	*Faecalibacterium* [35], *Ruminococcus* [35], *Bacteroides* [34]
Decreased abundance associated with decreased GVHD	NIL
Increase abundance associated with decreased GVHD	*Blautia* [19,21■,38,40], Lachnospiraceae [6■,21■,33], *Bacteroides* [21■,33,35]

GVHD, graft-versus-host disease.

of donor–recipient microbiota closeness or where novel GVHD treatment might include donor stool restoration [33]. Although alpha diversity can decline by as much as 30–50% peri-transplant, it has not consistently been associated with overall survival [9,19,32,34,35] (Table 1). This is at least partially explained by the heterogenous cohorts included in earlier allogeneic stem cell transplant microbiome studies with appreciation in retrospect that considerable variability exists between the microbiome of a heavily pretreated patient with acute myeloid leukaemia undergoing transplant compared with a patient undergoing curative transplant for a primary immunodeficiency syndrome without previous exposure to chemotherapy [41–44]. As the field of metagenomics and computational analysis has advanced, microbiota diversity is increasingly recognized to be a relatively crude representation of microbiota membership and its functional potential [12]. More important would appear to be the abundance of distinct commensal obligate anaerobes and their microbial metabolites or SCFAs, which interact with the host immune system although several distinct bacterial species have been positively or negatively correlated with subsequent GVHD or infectious outcome after allogeneic stem cell transplant [45].

The effect of antibiotics

The spectrum of activity, duration and timing of antibiotics are all independently important factors and variably influence the microbial ecosystem [13■]. Antibiotics can mediate long-lasting impacts and use up to 3 months prior to conditioning is associated with persistent disruptions in alpha diversity that remains detectable at the time of transplant [9]. Peri-transplant antianaerobic antibiotic usage is associated with reduced SCFA-producing enteric microbiota and reduced concentrations of luminal SCFA measurable in the 14 days posttransplant, both of which have been associated with a subsequent increased rate of GVHD-related mortality [28,46■,47]. Timing of antibiotic administration peri-transplant also has an important influence on the microbiome. When antibiotic use in 621 allogeneic stem cell recipients was stratified into early (administered between day −7 to day 0), late (administered after day 0) or none during transplant admission, early antibiotics were associated with reduced commensal anaerobic microbiota and greater treatment-related mortality after transplant because of GVHD [48]. Antibiotic exposure during transplantation is also associated with an increase in abundance of antimicrobial resistance gene expression, which in leukaemia cohorts is associated with reduced long-term overall survival [49,50]. Prophylactic antibiotics are also not without effect on the microbiome, and particular regimens have been associated with variable secondary effects including Enterococcal species expansion, resistance to the effects of subsequent systemic antibiotics, preservation of commensal anaerobes and GVHD mortality [13■,48,51–53]. Both therapeutic and prophylactic antibiotics have a profound effect on the microbiota and present a challenging confounder in transplant microbiome studies where they remain an intrinsic component of clinical care.

Infection

Loss of alpha diversity peri-engraftment is associated with subsequent *Clostridium difficile* colitis and expansion of enteric pathogens has preceded bacteraemia and extra-intestinal infection by the corresponding invasive Gram-positive and Gram-negative pathogen in adult and paediatric stem cell transplant recipients [13■,32,44,54,55]. Faecal metabolites measured in stool at the time of engraftment are associated with subsequent blood stream infection with low level of butyrate predictive of bacteraemia 30 days after transplant [36]. Postengraftment microbiome domination, where at least 30% of the microbiota is occupied by a single predominating bacterial taxon, is an additional predictor of subsequent bacteraemia [32]. A microbiome dominated by *Enterococcus* spp. has been associated with a three-fold increased risk of subsequent vancomycin-resistant enterococcal bacteraemia whereas a protobacteria-dominated microbiome has been associated with a five-fold increased risk of invasive Gram-negative bacteraemia. The microbiome has been predictive of all-cause infectious events posttransplant and in an Italian series of 96 patients, the presence of above 5% Enterobacteriaceae in stool at day −6 was predictive of subsequent sepsis whereas 10% or less Lachnospiraceae was an independent predictor for posttransplant infection-related death [9]. Recent studies suggest that the gut microbiota can affect the pulmonary inflammatory response and associated lung injury including in transplantation. Poststem cell transplant proteobacterial domination is associated with subsequent bacterial, viral and fungal respiratory tract infection and stratification of peri-engraftment microbiota based on SCFA production capability is an independent predictor of viral lower respiratory tract viral infection [46▪,56]. The broad effects of the gut–lung microbiome in transplantation are yet to be extensively studied but this expanding body of evidence would suggest that this bacterial immunological axis warrants further investigation.

FUTURE DIRECTIONS

Now that a large multinational, multisite, time-series cohort study has confirmed that features of the microbiome correlate with posttransplant outcomes, microbiome studies have turned to focusing on defining the molecular mechanisms by which these interactions occur [5▪▪]. A number of recent publications have incorporated additional methodologies to investigate these microbiome–immune interactions. Ha *et al.* [37] categorized a cohort of 89 allogeneic stem cell transplant recipients according to grade of GVHD and found the gut microbiota interrogated by 16S metagenomic sequencing was correlated with the balance of Treg/Th17 at the time of engraftment. Acute GVHD was associated with reduced diversity and depletion of anaerobic Clostridia, in particular Lachnospiraceae and Ruminococcaceae, and expansion of Gammaproteobacteria, including Enterobacteriaceaea. The abundance of Lachnospiraceae and Ruminococcaceae was positively correlated with Treg/Th17 ratio in the peripheral blood and negatively correlated with beta-lactam antibiotics and Enterobacteriaceaea. Adding an additional 69 cases to the initial cohort, now across two hospital centres with methodological strength of a discovery and validation study design, Han *et al.* describe the now 150-strong cohort with 16S metagenomic sequencing, peripheral multiplex immunoassays for cytokines and peripheral blood lymphocyte subset data [6▪]. GVHD-associated microbiota were again found to correlate with Treg/Th17 balance and an elevation in pro-inflammatory cytokines (including IL-1B, IL-6, IL-17A and TNF-a) providing further evidence to support the theory that the microbiota modulate GVHD outcome via enteric mucosal T-cell populations.

Romick-Rosendale *et al.* detailed a cohort of 42 paediatric allogeneic stem cell transplant recipients where posttransplant loss of diversity measured by 16S metagenomic sequencing was correlated with loss of butryrate-producing microbiota and increased GVHD. Metabolomics in serial stool samples confirmed a reduction in faecal butyrate paralleled a loss of butryrate-producing microbiota and both were predictive for those that went on to develop subsequent GVHD [46▪]. Four children underwent colonic biopsy during the study period and each had a tissue sample that was interrogated for RNA sequencing for the expression of known luminal butyrate transporters and receptors. Although no single target modified gene was identified this should not deter further research attempts to better define the pathway by which butyrate modulates the epithelium in GVHD. The correlation between diversity, butyrogenic microbiota, faecal butyrate and risk for GVHD has been widely observed in other cohorts and warrants further investigation [36].

CONCLUSION

It has become increasingly clear that the microbiome is a critical factor in maintaining human health in addition to disease pathogenesis. The microbiota offers significant potential as a source of novel diagnostic and therapeutic opportunities including in the immune-compromised, and in particular, recipients of allogeneic transplantation. Early experience with microbiota-based replenishment therapy in stem cell transplant provides evidence of diversity restoration and improvement in infectious and immunological outcomes. Furthermore, other recent publications hint at additional complimentary approaches that might include enteral nutrition [57–60]. Although the fields of metagenomics and computational biology continue to advance and a reduction in the once prohibitive cost improves the accessibility of these technologies, it becomes increasingly important that there is international consensus on standardizations for experimental design to ensure robust scientific output if the field is to make strides toward meaningful therapeutic interventions.

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Conflicts of interest

There are no conflicts of interest.

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